

Wilson-line Correlators in Perturbative QCD

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Based on 1909.00697 with Giulio Falcioni and Einan Gardi
and unfinished work to be completed

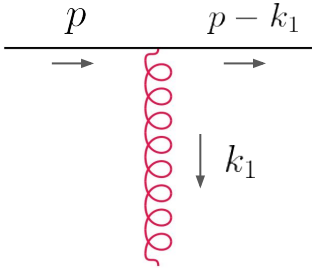
Torino [virtually :(]
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“introducing myself”

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[1909.00697]
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[20xx.xxxxx]
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[21xx.xxxxx]?

Eikonal approximation



$$= \frac{(\not{p} - \not{k}_1)\gamma^\mu}{(p - k_1)^2} \quad k_1 \rightarrow 0 \quad \text{soft limit}$$

$$\frac{\not{p}\gamma^\mu}{-2p \cdot k_1} \propto \frac{p^\mu}{p \cdot k_1} \quad \text{eikonal factor}$$

Eikonal factor is exactly the same as one from

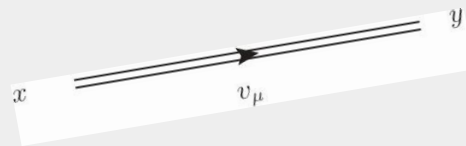
$$\int_0^\infty ds p^\mu A_\mu(sp) = \int_0^\infty ds p^\mu \int \frac{d^d k_1}{(2\pi)^d} \tilde{A}_\mu(k_1) e^{-isp \cdot k_1} \propto \int \frac{d^d k_1}{(2\pi)^d} \tilde{A}_\mu(k_1) \frac{p^\mu}{p \cdot k_1}$$

Multiple soft gluon emissions $\int_0^\infty ds p^\mu A_\mu(sp) \rightarrow \exp \left(\int_0^\infty ds p^\mu A_\mu(sp) \right)$

Wilson lines

A **Wilson line** in direction v between spacetime points x and y

$$W_v(x, y) = \mathbf{P} \exp \left(i g_s \int_x^y d^d z_\mu A^\mu(z) \right)$$



- These “mimic” the **soft gluon emissions** from a radiative particle $p \sim Qv$
- Direction is the **classical trajectory** of the particle since soft gluon emissions will not cause the particle to recoil
- If it is a massive particle then it is given by a **non-lightlike** line $v^2 \neq 0$
- If it is a massless particle then it is given by a **lightlike** line $v^2 = 0$
- Wilson lines are **spin-independent**
- They depend on the **colour representation** of the underlying particle (fundamental for quarks, adjoint for gluons etc.)

Wilson-line Correlators

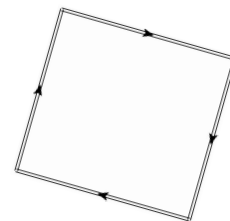
Under a local gauge transformation

$$W_\beta(y, x) \rightarrow e^{i\alpha^a(y)\mathbf{T}^a} W_\beta(y, x) e^{-i\alpha^a(x)\mathbf{T}^a} \quad A(x) = \mathbf{T}^a A^a(x)$$

Need to form **closed loops** to ensure gauge invariance

- Either forming n-gons

$$\text{tr} [W_{v_1}(x_1, x_2) W_{v_2}(x_2, x_3) \dots W_{v_n}(x_n, x_1)]$$



parallelogram or box

$W_\square$

- Or closing at infinity

$$W_{v_1}(\infty, 0) W_{v_2}(0, \infty)$$



wedge configuration

a pair of semi infinite lines $W_\wedge$

Eikonal functions
are made only from
Wilson lines

- Or more generally

$$W_{v_1}(\infty, 0) W_{v_2}(\infty, 0) \dots W_{v_n}(\infty, 0)$$

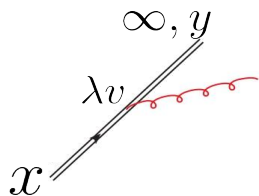
still gauge invariant
if **open colour indices** are at infinity

Computing Wilson-line Correlators

Convenient to work in **configuration** (coordinate) space

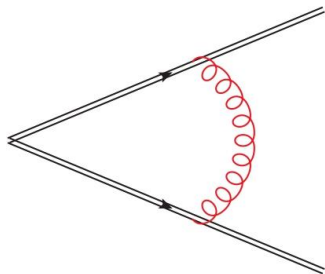


$$-\frac{\Gamma(1-\epsilon)}{4\pi^{2-\epsilon}} \frac{1}{(-x^2 + i0)^{1-\epsilon}} g_{\mu\nu}$$



$$ig_s \mathbf{T}^a v^\mu \int d^d z \int_0^\infty d\lambda \delta^d(z - x - \lambda v)$$

$$\int_0^1 d\lambda \delta^d(z - x - (y - x)\lambda)$$



Example The wedge at one-loop

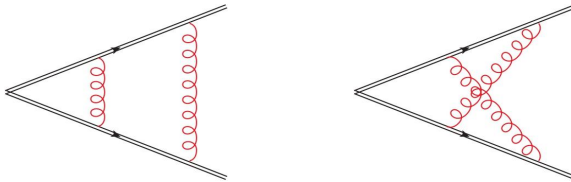
$$(ig_s)^2 \left(\frac{\Gamma(1-\epsilon)}{4\pi^{2-\epsilon}} \right)^2 \mathbf{T}_1^a \mathbf{T}_2^a \int_0^\infty \frac{d\lambda_1 d\lambda_2 v_1 \cdot v_2}{(-(\lambda_1 v_1 - \lambda_2 v_2)^2)^{1-\epsilon}}$$

for lightlike lines $\propto \int_0^\infty \frac{d\lambda_1}{\lambda_1^{1-\epsilon}} \frac{d\lambda_2}{\lambda_2^{1-\epsilon}}$ **Scaleless integral** $\Rightarrow$ cancellation between UV and IR

Diagrammatic Exponentiation

An (important) property of Wilson-line correlators is that they **exponentiate**

Two loop diagrams



$$\begin{aligned} &\propto g_s^4 \int_0^x d\lambda_1 d\lambda_2 d\sigma_1 d\sigma_2 \left(\theta(\lambda_1 - \sigma_1) \theta(\lambda_2 - \sigma_2) + \theta(\sigma_1 - \lambda_1) \theta(\lambda_2 - \sigma_2) \right) \\ &= g_s^4 \int_0^x d\lambda_1 \lambda_2 \sigma_1 \sigma_2 \theta(\lambda_2 - \sigma_2) = \frac{1}{2} \left(g_s^2 \int_0^x d\lambda_1 \lambda_2 D(\lambda_1 v_1 - \lambda_2 v_2) \right)^2 \end{aligned}$$

$$\text{correlator} = \text{triangle with 1 loop} + \text{triangle with 2 loops} + \text{triangle with 3 loops} = \text{triangle with 1 loop} + \frac{1}{2} \left(\text{triangle with 1 loop} \right)^2$$

This is for an **Abelian** gauge theory (QED)

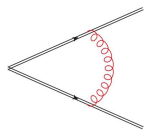
It can be shown that the correlator is given by the **exponential** of the one-loop graph to **all loop orders** (in the absence of matter)

$$= \exp \left(\text{triangle with 1 loop} \right)$$

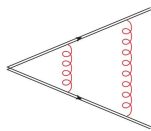
Diagrammatic Exponentiation

non-Abelian Exponentiation [Gatheral '83; Frenkel, Taylor '84]
Need to account for **colour factors**

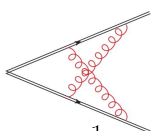
quadratic Casimir $C_i \equiv \mathbf{T}_1^a \mathbf{T}_2^a$



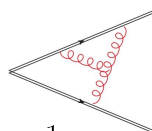
C_i



C_i^2



$C_i^2 - \frac{1}{2}C_i C_A$



$-\frac{1}{2}C_i C_A$

← “genuine” two loop contributions

$$\text{correlator} = \exp \left(C_i \text{ (triangle)} - \frac{1}{2}C_i C_A \text{ (triangle with self-energy)} - \frac{1}{2}C_i C_A \text{ (triangle with gluon self-energy)} \right)$$

Proportional to C_i
Casimir scaling extends to three loops
generalised version beyond

Keep only **maximally non-Abelian** colour factors in exponent
Diagrams in the exponent are called **webs**

Generalises to multiple Wilson lines [Gardi, Smilie, White '13]

Generating functional approach [Vladimirov's talk last week]

Multiplicative Renormalisation

Wilson loops have **UV divergences** which require renormalisation

non-lightlike [Korchemsky, Radyushkin '87]

RG for the **exponent** of the correlator $\Leftrightarrow$ computing **webs** gives direct access to **anomalous dimension**

$$\mu \frac{d}{d\mu} \log W_i = \Gamma_i$$

Labels the correlator $\nearrow$ will depend on kinematics $\nwarrow$

$$\lim_{\beta_i^2 \rightarrow 0} \Gamma_i \sim \gamma_{\text{cusp}} \log \beta_i^2$$

Universal lightlike cusp anomalous dimension $\nwarrow$

Lightlike lines have additional divergences [Korchemskaya, Korchemsky '92]

$$\mu \frac{d}{d\mu} \log W_i = \sum_{\text{cusps}} \gamma_{\text{cusp}} \log \mu + \Gamma_i$$

Subleading, geometry dependent (we will see this) $\nwarrow$

Now some **examples...**

Parallelogram Configuration

Two **lightlike** lines of **finite** length

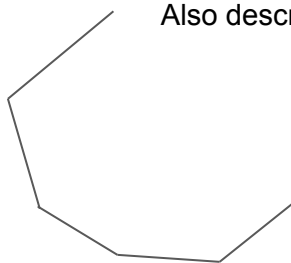
$$\mu \frac{d}{d\mu} \log W_{\square} = -2\gamma_{\text{cusp}} \left[\log(\mu^2 xy \beta_1 \cdot \beta_2) + \log(-\mu^2 xy \beta_1 \cdot \beta_2) \right] - \Gamma_{\square}$$

Two loop computation in [Korchemskaia, Korchemsky '92]

$$\gamma_{\text{cusp}} = \frac{\alpha_s}{\pi} C_i + \left(\frac{\alpha_s}{\pi} \right)^2 C_i \left[C_A \left(\frac{67}{36} - \zeta_2 \right) - \frac{5}{9} n_f T_f \right]$$

Casimir scaling

$$\Gamma_{\square} = \left(\frac{\alpha_s}{\pi} \right)^2 C_i \left[C_A \left(-\frac{11}{6} \zeta_2 - 7\zeta_3 + \frac{202}{27} \right) + n_f T_f \left(\frac{2}{3} \zeta_2 - \frac{56}{27} \right) \right]$$

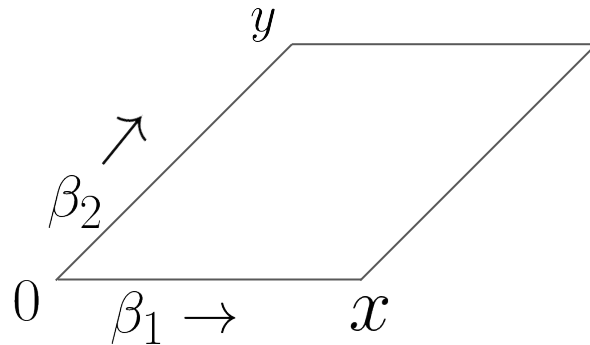


Also describes n-gon of multiple finite lightlike lines

[Drummond, Henn, Korchemsky and Sokatchev '09]

$$- \sum_{\text{cusps}} \gamma_{\text{cusp}} \log(\mu^2 x_i \cdot x_j) - \frac{n}{4} \Gamma_{\square}$$

(we will extend this picture to include infinite lines)



Wilson-loop Duality

[Brandhuber, Heslop, Travaglini '07; Drummond, Henn, Korchemsky and Sokatchev '08]

Such a configuration gives the finite terms in planar $\mathcal{N} = 4$ SYM amplitudes

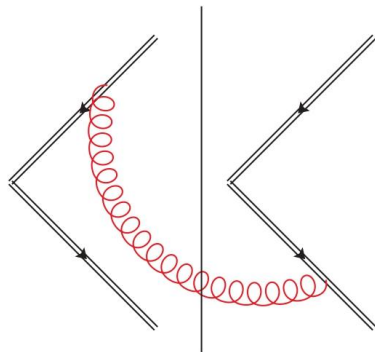
Why does it not describe also the IR divergent terms? (we will see this)

Drell-Yan Configuration

DY factorisation near threshold gives (very schematically)

$$Q^2 \frac{d\sigma_{\text{DY}}}{dQ^2} \sim \underset{\substack{\text{QCD form factor} \\ \sim \frac{\gamma_{\text{cusp}}}{\epsilon^2} + \frac{\gamma_G}{\epsilon}}}{\text{hard}} \times \underset{\substack{\text{Evolution (splitting functions) in} \\ \text{Mellin space behave as} \\ \sim -\gamma_{\text{cusp}} \log N + B_\delta}}{\text{PDF}} \times W_{\text{DY}} \longrightarrow$$

[Sterman '87; Catani, Trentadue '89; Korchemsky, Marchesini '93]



$$\mu \frac{dW_{\text{DY}}}{d\mu} = 2\gamma_{\text{cusp}} \log(x^2 \mu^2) + \Gamma_{\text{DY}} \quad \Gamma_{\text{DY}} = 2 \left(\frac{\alpha_s}{\pi} \right)^2 C_i \left[C_A \left(-\frac{11}{6} \zeta_2 - 7\zeta_3 + \frac{202}{27} \right) + n_f T_f \left(\frac{2}{3} \zeta_2 - \frac{56}{27} \right) \right]$$

[Belitsky '98]

Analysing **single poles** we get $\gamma_G - 2B_\delta - \frac{1}{2}\Gamma_{\text{DY}} = 0$ and by just comparing the two $\Gamma_{\square} = 2\Gamma_{\text{DY}}$
but finite terms are different

$$\gamma_G - 2B_\delta = \frac{\Gamma_{\square}}{4} \quad \text{spin dependence cancels}$$

Can we understand this better by looking at the component parts?

Wedge Configuration

Appears in the **infrared factorisation** of scattering amplitudes ^[Collins '89]

$$\mathcal{M}_n \sim \left(\prod_{i=1}^n \frac{J_i}{\mathcal{J}_i} \right) \mathcal{S}_n \mathcal{H}_n$$

collinear divergences
hard function

soft function

$\mathcal{S}_n \mathcal{J}_i J_i$
 have operator definitions and are highly constrained ^[Gardi, Magnea '09]

The soft function is described by semi-infinite lightlike Wilson lines

$$W_{v_1}(\infty, 0) W_{v_2}(\infty, 0) \dots W_{v_n}(\infty, 0)$$

Right now we shall special case massless and two legs (form factor)

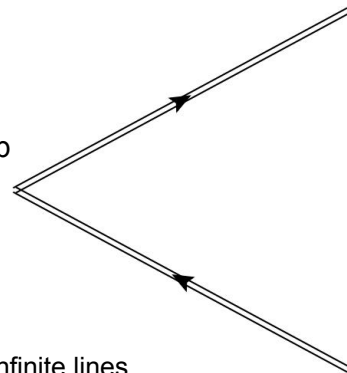
$$\mu \frac{d}{d\mu} \log W_\Lambda = -\gamma_{\text{cusp}} \log(\mu^2) - \Gamma_\Lambda$$

Lightlike semi-infinite lines notoriously difficult to disentangle UV from IR behaviour especially beyond one-loop

Two loop calculation performed in ^[Erdogan, Sterman '15]

$$\Gamma_\Lambda = \left(\frac{\alpha_s}{\pi} \right)^2 \frac{C_i}{4} \left[C_A \left(-\frac{11}{6} \zeta_2 - \zeta_3 + \frac{202}{27} \right) + n_f T_f \left(\frac{2}{3} \zeta_2 - \frac{56}{27} \right) \right]$$

Different value compared to $\Gamma_\square$ suggests a characterisation of the subleading AD in terms of finite/infinite lines



Wedge Configuration

Using the factorisation $\mathcal{M}_n \sim \left(\prod_{i=1}^n \frac{J_i}{\mathcal{I}_i} \right) \mathcal{S}_n \mathcal{H}_n$ and the constraint equations in [Gardi, Magnea '09]

$$\gamma_G = 2\gamma_{J/\mathcal{I}} - \Gamma_{\wedge}$$

single IR pole hard-collinear soft

Decomposes into spin-dependent and spin-independent terms

The quantity $\Gamma_{\wedge}$ also appears in the two-loop **gluon Regge trajectory**
 [Fadin, Kotsky, Fiore '95; Blumlein, Ravindran, van Neerven '98; Erdogan, Sterman '15]

At high energies amplitudes are dominated by t-channel exchanges and “reggeise”

$$\frac{1}{t} \rightarrow \frac{1}{t} \left(\frac{s}{-t} \right)^{\alpha(t, \epsilon)}$$

$$\alpha(t, \epsilon) = \frac{\alpha_s}{\pi} \frac{\gamma_{\text{cusp}}^{g(1)}}{2\epsilon} + \left(\frac{\alpha_s}{\pi} \right)^2 \frac{1}{4} \left(-\frac{C_A \hat{b}_0}{\epsilon^2} + \frac{\gamma_{\text{cusp}}^{g(2)}}{\epsilon} + 2\Gamma_{\wedge}^{g(2)} + C_A \hat{b}_0 \zeta_2 \right) + \mathcal{O}(\alpha_s^3)$$

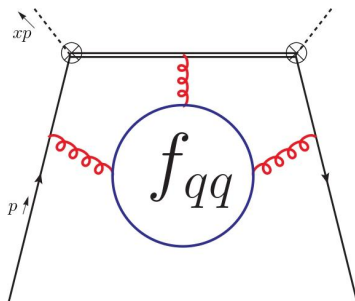
To the experts: at three loops it becomes **ambiguous** to define due to three-Reggeon exchanges

Perhaps understanding why $\Gamma_{\wedge}$ appears here will give a “nice” definition of the Regge trajectory

[Ridgway's talk at SCET 2020]

PDF Factorisation

We have decomposed this $\longrightarrow \gamma_G - 2B_\delta = \frac{\Gamma_\square}{4}$
 Now we look at doing the same on



Two partonic fields separated by **lightlike** Wilson line

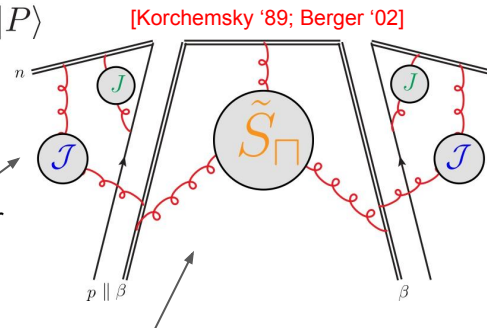
$$f_{qP}^{\text{bare}}(x, \epsilon) = \frac{1}{2} \int \frac{dy}{2\pi} e^{-iyxp \cdot u} \langle P | \bar{\psi}_q(yu) \gamma \cdot u W_u(y, 0) \psi_q(0) | P \rangle$$

[Collins, Soper '82]

The renormalised PDFs factorise in the limit of **soft gluon radiation**

$$\tilde{f}(N, \mu) = H \left(\frac{(2p \cdot n)^2}{n^2 \mu^2}, \alpha_s(\mu^2), \epsilon \right) \prod_{i=L,R} \frac{J_i \left(\frac{(2p \cdot n)^2}{n^2 \mu^2}, \alpha_s(\mu^2), \epsilon \right)}{\mathcal{J}_i \left(\frac{(2\beta \cdot n)^2}{n^2}, \alpha_s(\mu^2), \epsilon \right)} \times \tilde{S}_\square \left(N, \frac{\beta \cdot u \mu}{p \cdot u}, \alpha_s(\mu^2), \epsilon \right) + \mathcal{O} \left(\frac{\log N}{N} \right)$$

Same **collinear** behaviour as form factor



Different Wilson loop describing **soft** divergences

Bare PDFs are **scaleless**. The renormalised PDFs are purely IR poles

$$\tilde{f}(N, \mu) = \left(\prod_{i=L,R} \frac{J_i \left(\frac{(2p \cdot n)^2}{n^2 \mu^2}, \alpha_s(\mu^2), \epsilon \right) \Big|_{\text{pole}}}{\mathcal{J}_i \left(\frac{(2\beta \cdot n)^2}{n^2}, \alpha_s(\mu^2), \epsilon \right)} \right) \tilde{S}_\square \left(N, \frac{\beta \cdot u \mu}{p \cdot u}, \alpha_s(\mu^2), \epsilon \right) + \mathcal{O} \left(\frac{\log N}{N} \right)$$

we have used the fact that the eikonal functions are pure poles

PDF Factorisation

Bare PDFs are **scaleless**. The renormalised PDFs are purely IR poles

We have decomposed this $\longrightarrow \gamma_G - 2B_\delta = \frac{\Gamma_\square}{4}$
 Now we look at doing the same on

$$\tilde{f}(N, \mu) = \left(\prod_{i=L,R} \frac{J_i \left(\frac{(2p \cdot n)^2}{n^2 \mu^2}, \alpha_s(\mu^2), \epsilon \right) \Big|_{\text{pole}}}{\mathcal{J}_i \left(\frac{(2\beta \cdot n)^2}{n^2}, \alpha_s(\mu^2), \epsilon \right)} \right) \tilde{S}_\square \left(N, \frac{\beta \cdot u \mu}{p \cdot u}, \alpha_s(\mu^2), \epsilon \right) + \mathcal{O} \left(\frac{\log N}{N} \right)$$

This means that we can write the bare PDFs as pure UV poles and pure IR poles

$$\tilde{f}^{\text{bare}}(N) = \underbrace{\tilde{Z}(N)^{-1}}_{\text{UV}} \underbrace{\left\{ \left(\prod_{i=L,R} \frac{J_i|_{\text{pole}}}{\mathcal{J}_i} \right) \tilde{S}_\square(N) + \mathcal{O} \left(\frac{\log N}{N} \right) \right\}}_{\text{IR}}$$

this will give us the **splitting functions**

$$\frac{d}{d \log \mu} f_{jk}(x, \mu) = 2 \sum_{j'} \int_x^1 \frac{dz}{z} \overset{\text{splitting functions}}{P_{jj'}(z, \alpha_s)} f_{j'k}(x/z, \mu)$$

The soft function obeys the RG equation

another subleading AD

$$\mu \frac{d}{d\mu} \log \tilde{S}_\square = -2\gamma_{\text{cusp}}(\alpha_s(\mu^2)) \log \left(N \mu \frac{\beta \cdot u}{\sqrt{2} p \cdot u} \right) - \Gamma_\square(\alpha_s(\mu^2)) - \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \gamma_{\text{cusp}}(\alpha_s(\lambda^2, \epsilon)) + \mathcal{O} \left(\frac{\log N}{N} \right).$$

Using the fact that the bare PDF does not depend on the scale

We find a similar decomposition for splitting functions

$$2B_\delta = \overset{\text{"virtual AD" coefficient of } \delta(1-x)}{2\gamma_{J/\mathcal{J}}} - \overset{\text{hard-collinear}}{\Gamma_\square} - \overset{\text{soft}}{\Gamma_\square}$$

Testing the Decomposition

We have derived $2B_\delta = 2\gamma_{J/\mathcal{J}} - \Gamma_\square$

↑ “virtual AD” coefficient of $\delta(1-x)$
↑ hard-collinear
↑ soft

We can extract $\gamma_{J/\mathcal{J}}$ from $\gamma_G = 2\gamma_{J/\mathcal{J}} - \Gamma_\square$ since we know all the other parts

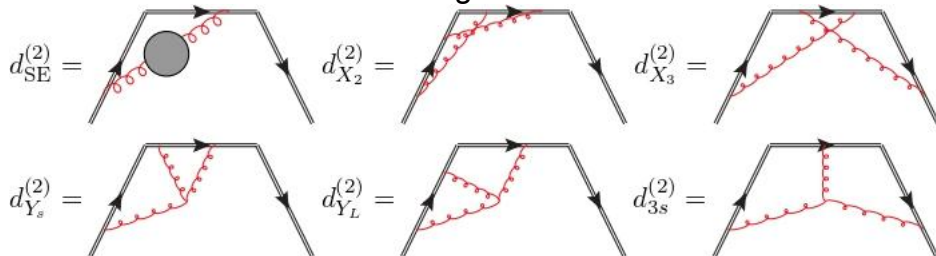
We know B_δ from explicit computation

We have a prediction for the two loop $\Gamma_\square$

$$\Gamma_\square = \left(\frac{\alpha_s}{\pi}\right)^2 \frac{C_i}{2} \left[C_A \left(-\frac{11}{6}\zeta_2 - 4\zeta_3 + \frac{202}{27} \right) + n_f T_f \left(\frac{2}{3}\zeta_2 - \frac{56}{27} \right) \right]$$

Simple check that it obeys Casimir scaling

In [Falcioni, Gardi, CM '19] we extend the calculation of two semi-infinite lines $W_\wedge$ to one with finite lines $W_\square$ without the need to use another regulator



By computing all the graphs we show that it is indeed true

Putting it together

$$\gamma_G = 2\gamma_{J/\mathcal{J}} - \Gamma_\wedge$$

form factor
decomposition

$$2B_\delta = 2\gamma_{J/\mathcal{J}} - \Gamma_\sqcap$$

share the same hard-collinear behaviour

$$\gamma_G - 2B_\delta = -\Gamma_\wedge + \Gamma_\sqcap$$

RHS can be seen as the
“eikonal version” of the LHS

but we know $\gamma_G - 2B_\delta = \frac{\Gamma_\square}{4}$

which implies

$$4(\Gamma_\sqcap - \Gamma_\wedge) = \Gamma_\square$$

Suggestive of **building block** picture

$$\Gamma_{\text{co}}^{\text{fin}} = \frac{1}{4}\Gamma_\square \quad \Gamma_{\text{co}}^{\text{inf}} = \frac{1}{2}\Gamma_\wedge$$

Difference is given by $\Gamma_{\text{co}}^{\text{fin}} = 2\Gamma_{\text{co}}^{\text{inf}} - \frac{3}{2}\left(\frac{\alpha_s}{\pi}\right)^2 C_i C_A \zeta_3 + \mathcal{O}(\alpha_s^3)$

Extends the idea from finite lines. For e.g.

$$\Gamma_\sqcap = 2\Gamma_{\text{infinite}} + \Gamma_{\text{finite}}$$

Maximal weight implies theory independent

$$\mathcal{N} = 4 \quad \text{or QCD}$$

Further work required: test with more geometries

Another interesting relation...

Quark/gluon field anomalous dimension in axial gauge γ_i gauge fixing condition $n \cdot A = 0$

Where the gluon propagator given by $\frac{i}{p^2} \left(-g^{\mu\nu} + \frac{n^\mu p^\nu}{n \cdot p} - \frac{n^\nu p^\mu}{(n \cdot p)^2} \right)$

$$\text{Wilson line becomes the identity} \\ W_n \sim \exp \left(\int d\lambda n \cdot A(\lambda n) \right) = 1$$

$B_\delta^i - \gamma_i = C_i(\dots)$ In [Korchemsky '89] it is shown that this difference obeys Casimir scaling

Main Goal: Can we prescribe a **Wilson loop geometry** to this difference?

A little known fact is that the **gluon field anomalous dimension** is given by the **beta function** [Kummer '74]

Allows for a prediction for the quark AD in axial gauge up to three loops $\gamma_q = \sum_n \left(\frac{\alpha_s}{4\pi} \right)^n \gamma_q^{(n)}$

$$\gamma_q^{(1)} = 3C_F \quad \gamma_q^{(2)} = C_F^2 \left(\frac{3}{2} - 12\zeta_2 + 24\zeta_3 \right) + C_A C_F \left(\frac{7}{2} + \frac{44}{3}\zeta_2 - 24\zeta_3 \right) + C_F T_f n_f \left(-2 - \frac{16}{3}\zeta_2 \right)$$

Intermediate Goal: Check $B_\delta^i - \gamma_i = C_i(\dots)$ and $\gamma_A = \beta$ by explicitly computing the axial gauge ADs

Computing the field ADs

- **Lightlike gauge vector** has been used before $n^2 = 0$
- Makes integrals easier
- But ambiguity arises about **additional collinear divergences** to n^μ
- Propose to calculate with non-lightlike gauge vector $n^2 \neq 0$

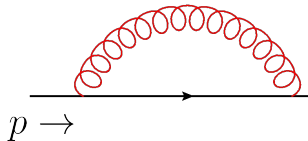
Jet function definition $\langle 0 | T [W_{n_i}(\infty, 0) \psi(0)] | p \rangle$
 Are the UV divergences the same as γ_q ?

$$\frac{i}{p^2} \left(-g^{\mu\nu} + \frac{n^{\{\mu} p^{\nu\}}}{n \cdot p} - \frac{n^2 p^\mu p^\nu}{(n \cdot p)^2} \right)$$

Linear propagator shares similarity with Wilson line integrals
 But this propagator lacks the $+i\epsilon$ prescription which is in the Wilson line propagator

One loop anomalous dimensions

Defining one-loop family $j_{a_1 a_2 a_3} = e^{\epsilon \gamma_E} \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{(k^2)^{a_1} ((p-k)^2)^{a_2} (k \cdot n)^{a_3}}$ ← complicated function of $\left(\frac{p \cdot n^2}{p^2 n^2}\right)$ evaluates to Meijer G-function



$$\frac{\alpha_s}{4\pi} (\epsilon - 1) (3j_{110} - 2p \cdot n j_{111})$$

← **finite** 2F1 function
 ← simple **divergent** bubble integral

$$\gamma_q^{(1)} = 3C_F$$

Gluon 1PI $\Pi^{\mu\nu} = \frac{\alpha_s b_0}{4\pi \epsilon} (p^2 \eta^{\mu\nu} - p^\mu p^\nu) + \mathcal{O}(\epsilon^0)$ still transverse to p^μ in this gauge

$$\gamma_A^{(1)} = b_0$$

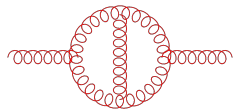
Two-loop non-lightlike axial gauge ADs

When you start looking into it you realise why it has not been done before

Follow a well-known flowchart for multi-loop computations with some difficulties for this specific problem

- Project onto the different **form factors** $\Pi^{\mu\nu} = G_1 \eta^{\mu\nu} + G_2 \frac{p^\mu p^\nu}{p^2} + G_3 \frac{n^{\{\mu} p^{\nu\}}}{n \cdot p} + G_4 \frac{n^\mu n^\nu}{n^2}$

- Multivariate partial fractioning** to project onto topologies



Gets complicated with diagrams like this
It has 5 gluon propagators each with $\sim \frac{1}{(p \cdot n)^2}$

→ able to map onto **three topologies**

$$\frac{p \cdot n}{((p - k_1) \cdot n)((p - k_2) \cdot n)((p - k_1 - k_2) \cdot n)} = \frac{1}{((p - k_1) \cdot n)((p - k_1 - k_2) \cdot n)} + \frac{1}{((p - k_2) \cdot n)((p - k_1 - k_2) \cdot n)} - \frac{1}{((p - k_1) \cdot n)((p - k_2) \cdot n)}$$

- IBP reduction to basis integrals: Kira and FIRE used to find **35 basis integrals** $\mathbf{f}(p^2, n^2, p \cdot n, \epsilon)$
- Derive **differential equations** for the integrals Can factor out dependence on p^2 and n^2 using dimensionality and scaling invariance
One “real” variable $x = \frac{p \cdot n}{\sqrt{p^2} \sqrt{n^2}}$ LiteRed $\partial_x \mathbf{f}(x, \epsilon) = A(x, \epsilon) \mathbf{f}(x, \epsilon)$ **Rationalise alphabet** by $x = \frac{z^2 - 1}{2z}$
- Transform to **canonical form** [Henn ‘11] $\mathbf{f} \rightarrow \mathbf{g}$ $\partial_x \mathbf{g}(x, \epsilon) = \epsilon \tilde{A}(x) \mathbf{g}(x, \epsilon)$ $\{z^2, 1 - z^2, 2 - z^2, 1 + z^2, 1 - 2z^2\}$
Multiple polylogarithms $G(a_1, \dots, a_n; z)$
- Boundary conditions** $p \cdot n \rightarrow 0, z \rightarrow 1$ use **Mellin-Barnes for asymptotic expansion**, difficult ones we fit numerically

Some tentative results and issues

Write the quark self-energy as $\Sigma = F_1 \not{p} + F_2 \frac{p \cdot n}{n^2} \not{n}$

Our result: $\gamma_q^{(2)} = C_F^2 \left(-\frac{5}{2} + 16\zeta_2 \right) + C_A C_F \left(\frac{49}{6} + 24\zeta_2 \right) + C_F T_f n_f \left(-\frac{5}{3} \right)$

- **Differs greatly** from the predicted result from $B_\delta^i - \gamma_i = C_i(\dots)$ $\gamma_A = \beta$
 $\gamma_q^{(2)} = C_F^2 \left(\frac{3}{2} - 12\zeta_2 + 24\zeta_3 \right) + C_A C_F \left(\frac{7}{2} + \frac{44}{3}\zeta_2 - 24\zeta_3 \right) + C_F T_f n_f \left(-2 - \frac{16}{3}\zeta_2 \right)$
- No dependence on z in the result. All those terms cancelled which is a **non-trivial check**.
 We could have specified z but chose not to.
- No term of **maximal weight** ζ_3 it would be surprising that $\gamma_A = \beta$ which does not have ζ_3 that γ_q would

Why are we **not totally convinced** by our result?

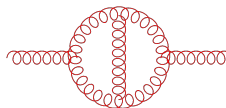
We also find $F_2 \sim \left(\frac{\alpha_s}{4\pi} \right) C_A C_F \frac{\zeta_2}{\epsilon}$ would suggest having to add a counterterm $\bar{\psi} \frac{\not{n} \cdot \partial}{n^2} \psi$

We have naively assumed the $+i\epsilon$ while computing boundary conditions

$$\frac{1}{p \cdot n} \rightarrow \frac{1}{2} \left[\frac{1}{p \cdot n + i\epsilon} + \frac{1}{p \cdot n - i\epsilon} \right]$$

will affect π^2
but not ζ_3

A **computation of the gluon** would also help solve issues



complicated diagram
just technical computer problems

Changing motivation to verifying $\gamma_A = \beta$

Non-lightlike semi-infinite Wilson lines

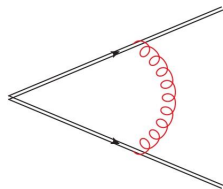
Describes **soft divergences in massive amplitudes**

$$\mathcal{M}_n \sim \mathcal{S}_n \mathcal{H}_n$$

$$\mathcal{S}_n \sim W_{v_1}(\infty, 0) W_{v_2}(\infty, 0) \dots W_{v_n}(\infty, 0)$$

$$p_i \sim m v_i$$

Understanding the **mathematical structure** of $\mathcal{S}_n$ will help us understand what can appear in $\mathcal{M}_n$
Can be thought of as a **testing ground**



One-loop computation

$$\alpha + \frac{1}{\alpha} = \frac{2v_1 \cdot v_2}{\sqrt{v_1^2} \sqrt{v_2^2}} \quad r(\alpha) = \frac{1 + \alpha^2}{1 - \alpha^2}$$

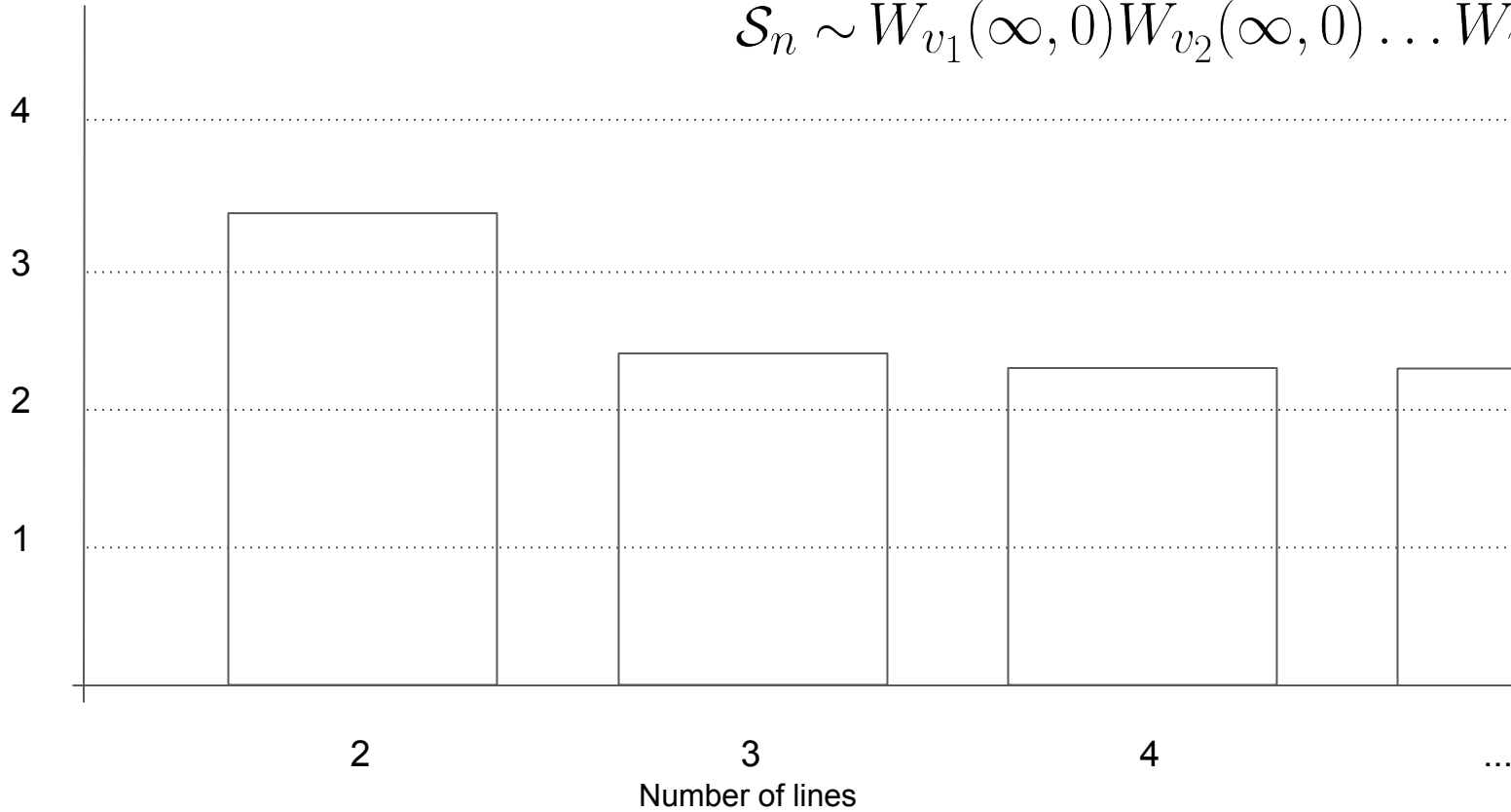
$$(ig_s)^2 \left(\frac{\Gamma(1-\epsilon)}{4\pi^{2-\epsilon}} \right)^2 \mathbf{T}_1^a \mathbf{T}_2^a \int_0^\infty \frac{d\lambda_1 d\lambda_2 v_1 \cdot v_2}{(-(\lambda_1 v_1 - \lambda_2 v_2)^2)^{1-\epsilon}} \sim r(\alpha) \log(\alpha) \left(\frac{1}{\epsilon_{UV}} - \frac{1}{\epsilon_{IR}} \right)$$

We are interested in IR divergences but we can trade the IR for the UV using the fact that the correlator is **scaleless**. Regulate the IR using an exponential cut-off etc.

Status of Calculations

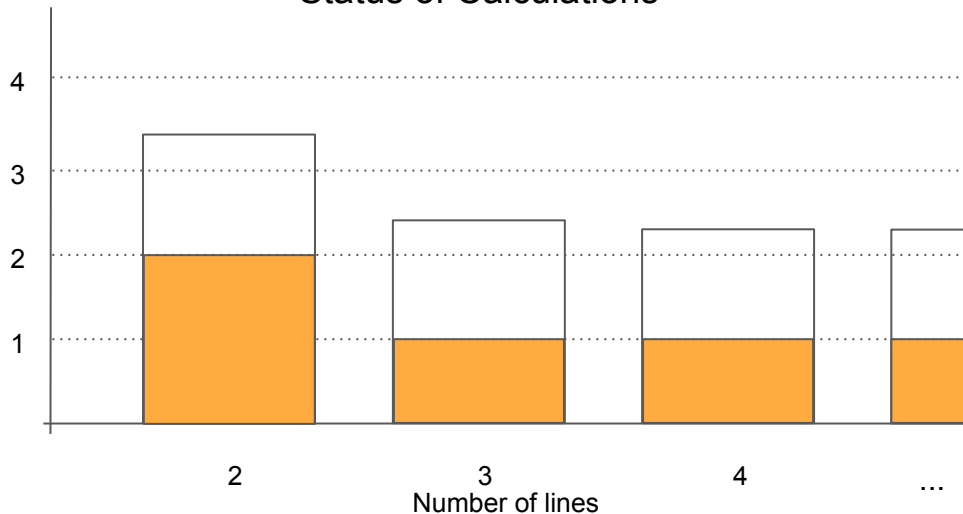
$$\mathcal{S}_n \sim W_{v_1}(\infty, 0) W_{v_2}(\infty, 0) \dots W_{v_n}(\infty, 0)$$

Number of loops



Number of loops

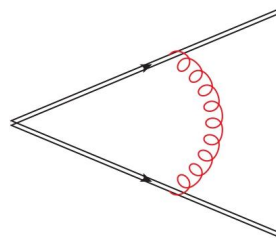
Status of Calculations



special two-line case

Two-loop cusp calculation

[Korchemsky, Radyushkin '87]



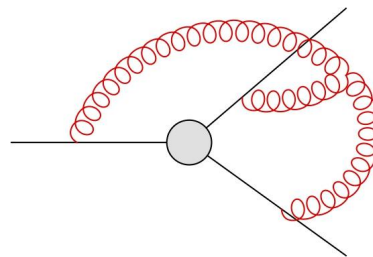
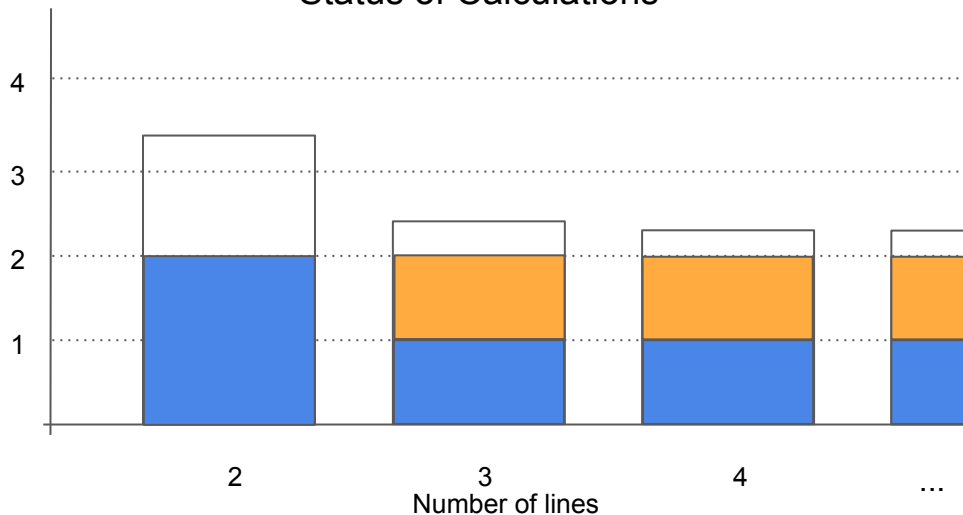
The one loop automatically gives result for any number of lines i.e. sum over pairs of the one-loop graph

The **lightlike cusp** can be found from the limit
$$\lim_{\alpha \rightarrow 0} \Gamma_{\text{cusp}}(\alpha) \sim \gamma_{\text{cusp}} \log(\alpha) + \text{constant}$$

Two-loop three-line calculation

Number of loops

Status of Calculations



Difficult but single poles gives very simple result

$$\sim \epsilon_{ijk} r(\alpha_{ij}) \log(\alpha_{ij}) \log^2(\alpha_{ik})$$

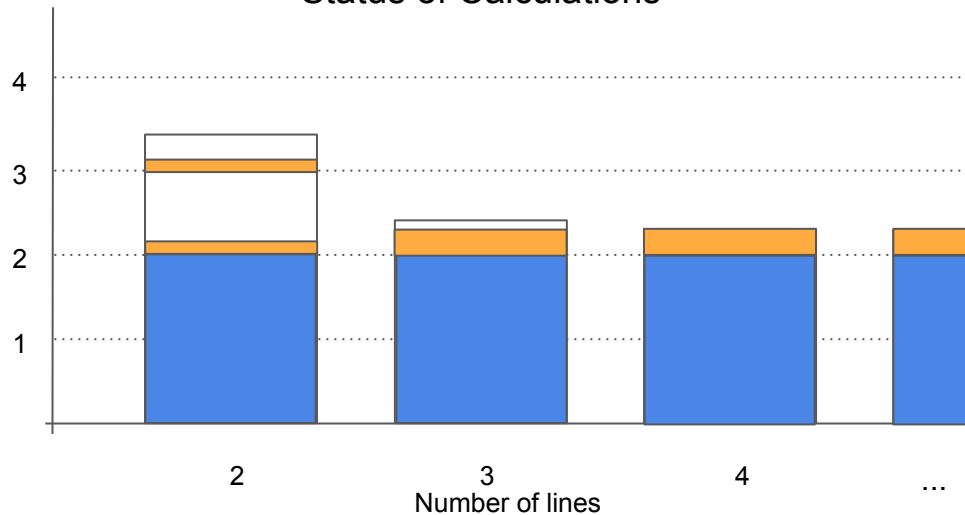
- Mellin-Barnes [Ferrogia, Neubert, Pecjak and Yang '09; Almelid '16]
- Unitarity cuts [Waelkens '17]
- Differential equations complicated, 6x6 coupled block
Maximal cut suggests elliptic

$$\sim \int \frac{dz}{\sqrt{(z-b_1)(z-b_2)(z-b_3)(z-b_4)}}$$

Dependent on regulator?

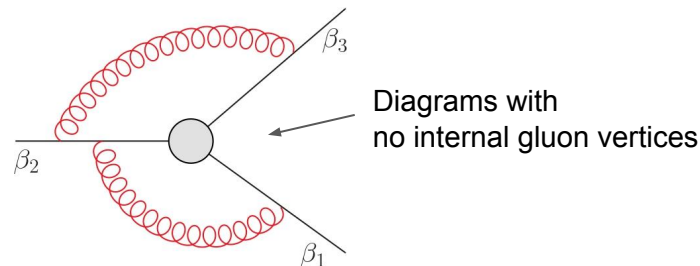
Number of loops

Status of Calculations



Multiple Gluon Exchange Webs

[Gardi '13; Falcioni, Gardi, Harley, Magnea, White '14]



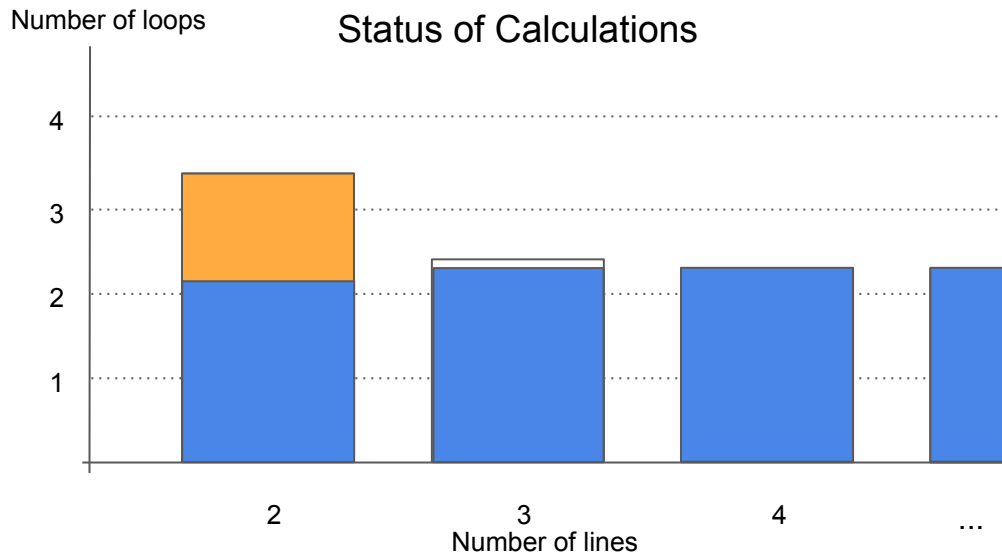
Although complicated integrals still have to be done, the **function space is well-understood**

Conjectured alphabet

$$\mathcal{A} = \left\{ \alpha, \eta \equiv \frac{\alpha}{1 - \alpha^2} \right\}$$

One rational function

$$r(\alpha) = \frac{1 + \alpha^2}{1 - \alpha^2}$$



Three-loop cusp calculation

[Grozin, Henn, Korchemsky, Marquard '15]

- **Differential equations method**
Using exponentiation to rewrite non-planar graphs in terms of planar graphs

New rational $r(\alpha) = \frac{1 + \alpha^2}{1 - \alpha^2}, s(\alpha) = \frac{\alpha}{1 - \alpha^2}$

New letter $\mathcal{A} = \{\alpha, \eta \equiv \frac{\alpha}{1 - \alpha^2}, y \equiv \frac{1 + \alpha}{1 - \alpha}\}$

Four-loop cusp calculation

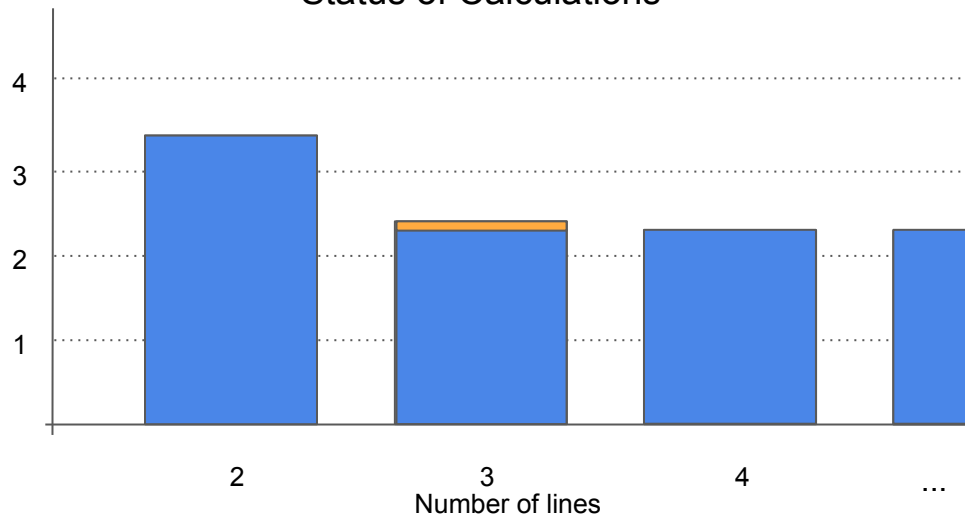
[Grozin '18; Bruser, Grozin, Henn, Stahlhofen '19]

- **Some colour terms known**
Using information on finite $\alpha \rightarrow 1$ limit can expand integrands prior to integration

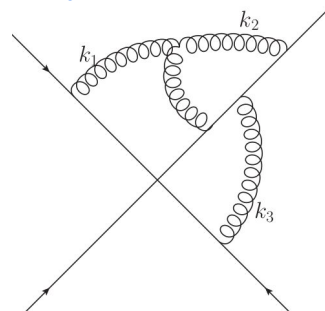
Three-loop three-line connected web

Number of loops

Status of Calculations



The **only one computed of this type** is using unitarity cuts



[Waelkens '17]

Function alphabet shared with MGEW type

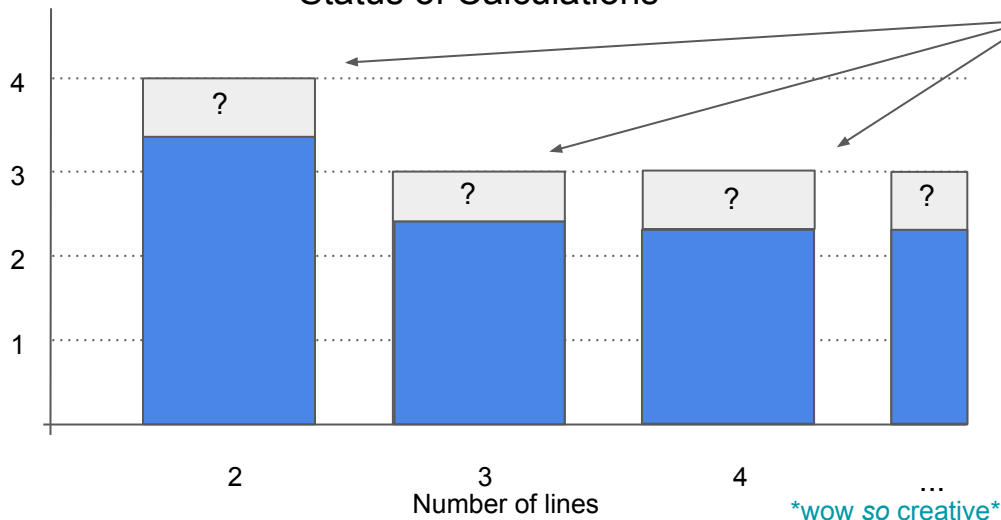
$$\mathcal{A} = \left\{ \alpha, \eta \equiv \frac{\alpha}{1 - \alpha^2} \right\}$$

$$r(\alpha) = \frac{1 + \alpha^2}{1 - \alpha^2}$$

Number of loops

Status of Calculations

How do we get **these** pieces?



Although great strides have been made in understanding the web structure of higher order terms **no significant progress** has been made recently with regards to the **functions** that appear in multi-line computations

[Agarwal, Danish, Magnea, Pal, Tripathi '20]

Bootstrapping

some *very* early stage work...

In the *lightlike case* a function space is known, an ansatz was constructed and constrained for the three-loop
 [Almelid, Duhr, Gardi, McLeod, White '17]

How about the non-lightlike case?

Take the known information and apply symmetries

$$r(\alpha) = \frac{1 + \alpha^2}{1 - \alpha^2}, s(\alpha) = \frac{\alpha}{1 - \alpha^2}$$

$$\alpha \rightarrow \frac{1}{\alpha}$$

$$\mathcal{A} = \left\{ \alpha, \eta \equiv \frac{\alpha}{1 - \alpha^2}, y \equiv \frac{1 + \alpha}{1 - \alpha} \right\}$$

$$\alpha \rightarrow -\alpha$$

Symbol invariance [Gardi '13]

		no. of α	
no. of y		odd	even
		$r(\alpha)^{\text{even}} s(\alpha)^{\text{odd}}$ $r(\alpha)^{\text{odd}}$	$r(\alpha)^{\text{odd}} s(\alpha)^{\text{odd}}$ $r(\alpha)^{\text{even}}$

These are functions for diagrams that *factorise the dependence on angles* $\sim f(\alpha_1)g(\alpha_2) \dots$

Will *NOT* be the case for all: the lightlike case tells us that *there is complicated dependence on multiple angles*

Conjecture that *only* diagrams that depend on CICRs in lightlike do not factorise on angles

$$\frac{(v_1 \cdot v_2)(v_3 \cdot v_4)}{(v_1 \cdot v_3)(v_2 \cdot v_4)}$$

Cusp Bootstrap?

some *very* early stage work...

1. Write ansatz for the cusp $\Gamma_{\text{cusp}}(\alpha) = \sum_i c_i f_i(\alpha)$

2. Initially constrain using the limits

$$\lim_{\alpha \rightarrow 0} \Gamma_{\text{cusp}}(\alpha) \sim \gamma_{\text{cusp}} \log(\alpha) + \text{constant}$$

known to four loops [Henn, Korchemsky, Mistlberger '19]

unknown but form of divergence known

[Bruser, Grozin, Henn, Stahlhofen '18]

$$\lim_{\alpha \rightarrow -1} \Gamma_{\text{cusp}}(\alpha) = \frac{V}{1+\alpha} + \mathcal{O}\left(\frac{\alpha_s^4 \log(1+\alpha)}{1+\alpha}\right)$$

3. Then constrain rest using $\alpha \rightarrow 1$ limit by expanding integrand and integrating

Loop	# of functions	How far in $\alpha \rightarrow 1$
1	2	Not needed
2	24	8th order
3	137	>20th order

Next steps:

- Limit the growth by understanding more about the possible functions
- Multiple lines?

Conclusion

Shown

$$\gamma_G - 2B_\delta = -\Gamma_\wedge + \Gamma_\sqcap$$

form factor single IR pole
coefficient of $\delta(1-x)$

subleading Wilson loop ADs

Conjectured

$$\Gamma_\sqcap = 2\Gamma_{\text{infinite}} + \Gamma_{\text{finite}}$$

Connection to Regge trajectory unknown

$$\alpha(t, \epsilon) = \frac{\alpha_s}{\pi} \frac{\gamma_{\text{cusp}}^{g(1)}}{2\epsilon} + \left(\frac{\alpha_s}{\pi}\right)^2 \frac{1}{4} \left(-\frac{C_A \hat{b}_0}{\epsilon^2} + \frac{\gamma_{\text{cusp}}^{g(2)}}{\epsilon} + 2\Gamma_\wedge^{g(2)} + C_A \hat{b}_0 \zeta_2 \right) + \mathcal{O}(\alpha_s^3)$$

Quark/gluon field anomalous dimension in **axial gauge** γ_i

Checking the relations $B_\delta^i - \gamma_i = C_i(\dots)$ and $\gamma_A = \beta$

General **function space** for non-lightlike lines

Conclusion

Shown

$$\underbrace{\gamma_G}_{\text{form factor single IR pole}} - 2\underbrace{B_\delta}_{\text{coefficient of } \delta(1-x)} = \underbrace{-\Gamma_\wedge}_{\text{subleading Wilson loop ADs}} + \Gamma_\sqcap$$

Conjectured

$$\Gamma_\sqcap = 2\Gamma_{\text{infinite}} + \Gamma_{\text{finite}}$$

Connection to Regge trajectory unknown

$$\alpha(t, \epsilon) = \frac{\alpha_s}{\pi} \frac{\gamma_{\text{cusp}}^{g(1)}}{2\epsilon} + \left(\frac{\alpha_s}{\pi}\right)^2 \frac{1}{4} \left(-\frac{C_A \hat{b}_0}{\epsilon^2} + \frac{\gamma_{\text{cusp}}^{g(2)}}{\epsilon} + 2\Gamma_\wedge^{g(2)} + C_A \hat{b}_0 \zeta_2 \right) + \mathcal{O}(\alpha_s^3)$$

Quark/gluon field anomalous dimension in **axial gauge** γ_i

Checking the relations $B_\delta^i - \gamma_i = C_i(\dots)$ and $\gamma_A = \beta$

General **function space** for non-lightlike lines

THANKS FOR LISTENING